

Name: **SOLUTIONS** ()Class: **26S0**

Date:

- 1 In the previous planning experiment, you established a method to record background radiation counts C at varying horizontal distances b from a building and vertical distances s from the soil using a Geiger-Müller (GM) tube.

To further investigate the contribution of soil radiation, an automated outdoors data-logging rig was set up at a fixed distance $b = 0.50$ m from the building wall. The detector was moved vertically to collect 500 readings of total background count rate C (in counts per minute, cpm) across various heights s (in meters, m) above the ground.

The data collected for s and C is recorded in the digital file: **Radiation_Data_dataset.xlsx**.

It is suggested that radiation originating from radioisotopes in the soil decays exponentially with height, while merging into a baseline background radiation count rate C_0 . The model for the total measured count rate C as a function of height s is

$$C = C_0 + Ae^{-ks}$$

where C_0 , A and k are constants.

- (a) Upon inspection, the data collected at large heights indicates that the baseline background radiation count rate stabilises at $C_0 \approx 18.00$ cpm.

- (i) Identify three specific data points that are anomalous.

Describe how you identified these points and how you should treat these points before further analysing the data.

The three data points are: (0.450, 68.13) in row 46, (1.800, 18.43) in row 181 and (3.500, 42.29) in row 351.

Plot a scatter plot of C against s using the data. The points that lie far from the trend of the other data are the anomalous points.

These points should be removed from the data set.

[2]

- (ii) Calculate $\ln(C - C_0)$ for all valid data points.

State and explain what happens to $\ln(C - C_0)$ for heights $s > 4.000$ m where background noise causes $C \leq C_0$.

For $C \leq 18.00$, $(C - C_0) \leq 0$ and $\ln(C - C_0)$ is not possible to calculate.

This happens because the actual count rate from the soil decrease to very small values at large heights such that the measured count rate is mostly due to the background count of 18.00 cpm.

[2]

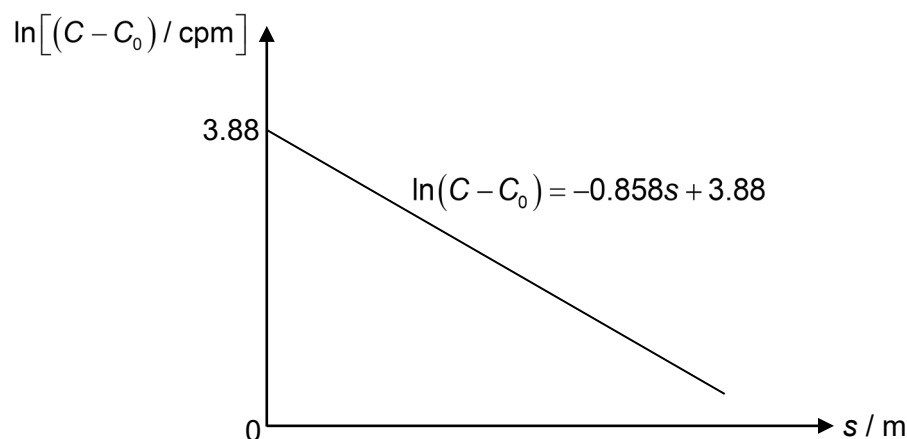
- (iii) Plot a suitable graph using a spreadsheet to determine the constants A and k .

Sketch the graph in the space below. Write down the equation of the trendline.

(Note: In your analysis, you should exclude the anomalous data points identified in (a)(i) and the data points where $s > 4.000$ m.)

$$\ln(C - C_0) = -ks + \ln A$$

Plot a graph of $\ln(C - C_0)$ against s where the gradient is $-k$ and the vertical intercept is $\ln A$.



$$\text{gradient} = -0.858$$

$$-k = -0.858$$

$$k = 0.858 \text{ m}^{-1}$$

$$\text{vertical intercept} = 3.88$$

$$\ln A = 3.88$$

$$A = e^{3.88} = 48.424 = 48.4 \text{ cpm}$$

$$A = 48.4 \text{ cpm}$$

$$k = 0.858 \text{ m}^{-1}$$

[3]

If exclude the 3 negative values of $\ln(C - C_0)$ [(3.830, -1.347), (3.930, -0.342), (3.970, -1.022)] as these lie far from the linear trend,

Trendline: $\ln(C - C_0) = -0.843s + 3.86$

$A = 46.5 \text{ cpm}$, $k = 0.843 \text{ m}^{-1}$

- (b) The spatial rate of change of count rate with height is given by the gradient of the count rate against height graph.

Determine the spatial rate of change of count rate with height at $s = 0.500 \text{ m}$.

For $s = 0.500 \text{ m}$ in row 50

$$\text{gradient} = \frac{B49 - B51}{A49 - A51} = \frac{49.91 - 49.36}{0.490 - 0.510} = -27.5$$

OR use the Excel function: =SLOPE(B49:B51,A49:A51)

$$\text{rate of change} = -27.5 \text{ cpm m}^{-1} \quad [1]$$

- (c) The total radiation exposure corresponds to the area under the count rate against height graph over this range.

- (i) Determine the total radiation exposure between heights $s = 0.100 \text{ m}$ and $s = 3.000 \text{ m}$.

For $s = 0.100 \text{ m}$ in row 11 and $s = 3.000 \text{ m}$ in row 299

Calculate area of trapezium in column D

Use the trapezoidal rule formula in Excel from row 12 to row 299 where the first value of i is 12: $=0.5 * (C_{i-1} + C_i) * (s_i - s_{i-1})$

Sum up the area of the trapeziums within the range: =SUM(D12:D299)

$$\text{radiation exposure} = 99.6 \text{ cpm m} \quad [1]$$

- (ii) State the primary source of experimental uncertainty in radioactive count rates that would have resulted in uncertainty in the answer in (c)(i).

Describe one method that would reduce this uncertainty.

The primary source of uncertainty is the fluctuation in the count rate due to the random nature of radioactive decay.

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Increase the time interval over which the counts are recorded at each height (e.g. from 1 min to 5 min). This will reduce the random error in the calculation of the counts per minute.
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OR

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Repeat multiple count readings at each height and calculate the average to reduce random error.
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..... [2]

(d) Air acts as a natural attenuating medium for soil radiation.

Determine the minimum height above the ground a worker must stand so that the air column provides at least 90 % reduction in soil radiation exposure compared to ground level.

$$C_{soil} = Ae^{-ks}$$

$$\frac{C_{soil}}{A} = e^{-ks}$$

$$\frac{10}{100} = e^{-ks}$$

$$\ln e^{-ks} = \ln \frac{10}{100}$$

$$-ks = \ln 0.10$$

$$s = \frac{\ln 0.10}{-k} = \frac{\ln 0.10}{-0.858} = 2.6837 = 2.68 \text{ m}$$

height = 2.68 m [1]